lowCortisol

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[Find My **OSHI** !!!]

It’s the big 26, and you don’t have a fav VTuber? Look no further! Take a quiz, find, favorite your Oshi, view their stats and make a post to share your opinion! Every VTuber will have a page displaying a photo of their model, a link to their channel, their subscriber count, video count, total/average likes/comments, etc. If they stream, data from their YouTube streams will be displayed, including information on their chatter data (how many unique chatters, unique messages) and their superchat data (unique superchats, revenue by superchats). Go and find your favorite Oshi!

COMPONENTS:

__init__.py:

Description:
- handles database connection
- allows for display of html pages

auth.py:

Description:
- handles login/register logic

download_datasets.py:

Description:
- Downloads datasets from kaggle into a designated dir

build_db.py:

Description:
- Creates database file and tables in the db file oshi.db
- Tables include those that store user data and vtuber data

loader.py:

Description:
- Inserts data from kaggle csvs into our db

home.html:

Description:

- homepage
- shows description of:
    - aim of the website
    - display of your select favorite VTubers

header.html:

Description:

- base for all other pages
- contains a header with links to:
    - Homepage
    - Github repo
    - Quiz
    - List of Vtubers
    - Login/Logout depending if you're logged in or not

login.html:

Description:

- allows users to log into their accounts
- visible link to register page
- throws error if:
    - account doesn't exist
    - incorrect password

register.html:

Description:

- allows users to create an account
- will throw error if:
    - username already exists in db
    - password response doesn't match with confirm password response

browse.html:

Description:

- Allows users to view all vtubers enlisted in the db
- Users can filter by agency
- Search function is also available if users have a specific vtuber they want to look at

profile.html:

Description:

- displays VTuber channel name, stats and profile picture
- option to favorite to display on home page
- 3 charts: Chat Activity Over Time, Member Data Over Time, Viewer Interaction Distribution (pie chart)
- has a section to post user's opinion on the VTuber
- Users can favorite the vtuber on their profile page

quiz.html:

Description:

- goal: to find your VTuber
- set of 5 handpicked questions that narrow down every aspect of an ideal vtuber
- Each question narrows down the amount of available vtubers

quiz_results.html:

Description:

- Shows the results from quiz.html
- Displays the top 5 vtubers matching the answers from the quiz, ordered by youtube subscriber count
- Displays error message if no vtubers match the answers from the quiz

update_pfp.py:

Description:

- Uses Holodex API to update profile pictures of outdated vtubers

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DATASETS USED:

VTuber 1B Elements: Live Chat Statistics - https://www.kaggle.com/datasets/uetchy/vtuber-livechat-elements

VTuber Data - https://www.kaggle.com/datasets/maliqr/vtuber-like-views-and-subscriber-data

API: https://holodex.net/login

DATABASES:

users		
string	username	PK
string	password	

vtubers		
string	channel_id	PK
string	channel_name	
string	profile_image_url	
string	channel_url	
string	agency	
int	total_subscriber_count	
int	total_videos	
int	total_views	
int	total_likes	
int	total_comments	
int	avg_views	
int	avg_likes	

chats		
string	channel_id	FK
string	period	
int	chats	
int	member_chats	
int	unique_chatters	
int	banned_chatters	
int	deleted_chats	

superchats		
string	channel_id	FK
string	period	
int	super_chats	
int	unique_super_chatters	
int	total_sc	
int	average_sc	
int	total_message_length	
int	average_message_length	
string	most_frequent_currency	
string	most_frequent_color	

comments		
int	comment_id	PK
string	channel_id	FK
string	username	FK
string	desc	

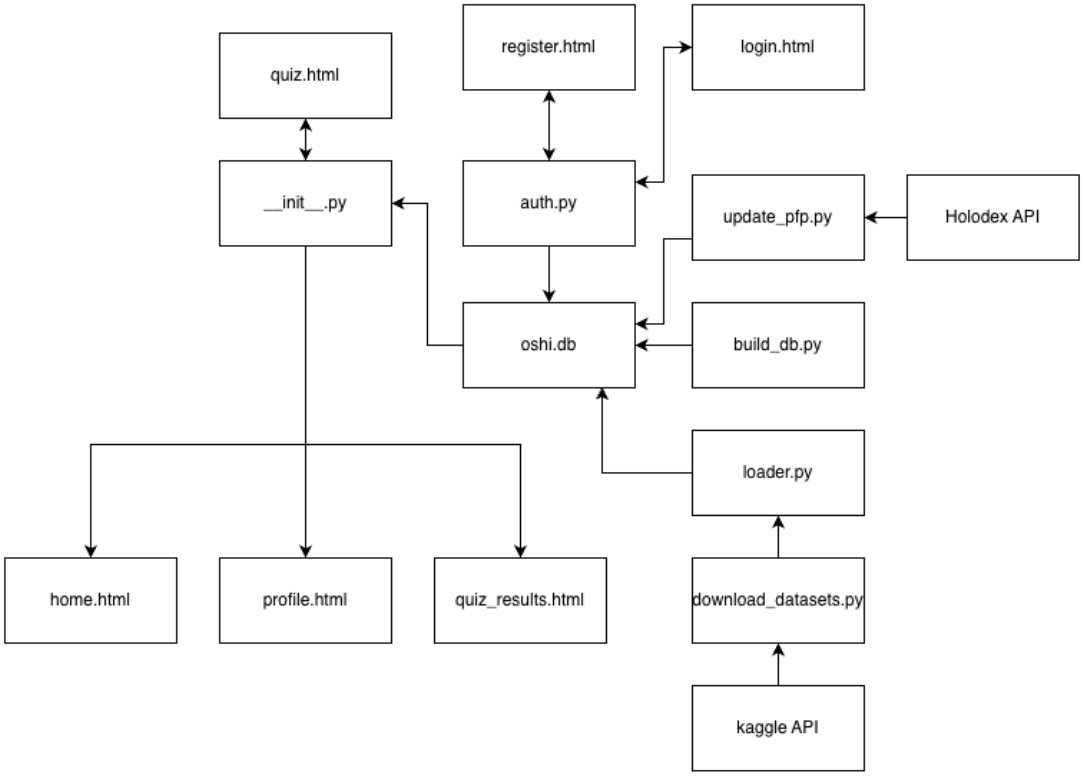
favorites		
string	username	FK
string	channel_id	FK

FRONT-END FRAMEWORK: We chose Tailwind because it has straightforward documentation and allows for dynamic, vibrant, and more styling.

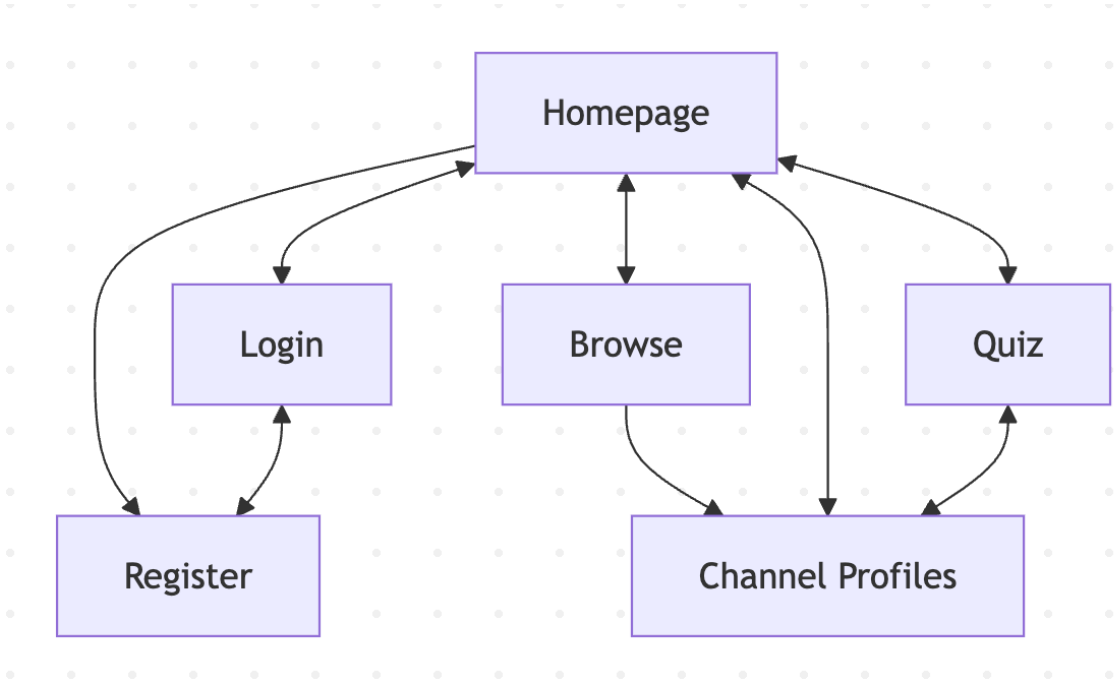
DATABASE: We chose to use a SQLite3 database because we are more familiar with SQL. Using SQLite3, we will be able to easily access and edit our database tables. SQL queries will also help in filtering certain data as needed.

DATABASE VISUALIZATION: We will be using chart.js because of the ease to create simple data visualization like pi chart, bar charts and graphs. This will help display basic stats but also data over time using the time periods. Chart.js is also inherently interactive, meaning that users can choose to show/hide specific inputs on graphs to better see the relationships between other inputs.

COMPONENT MAP:



SITE MAP:



TEAM ROLES:

Jun Jie
- Project manager. Make sure deliverable checkpoints are met.
- Handle Flask routing and SQLite queries
- Favorite VTubers button and list of them in home page

Shafin
- Visualize Data with charts
- Handle profile organization
- Handle kaggle dataset downloading logic
- Handle creation + population of db
- Create quiz
- Code search and filter functionalities across VTuber pages

Lucas
- Help create tables for visualization
- Store picture of tables in folders rather than constant generation

Kyle
- Handle templates and site design with Tailwind
- Link HTML with Jinja for conditional rendering of templates